UTKARSH DWIVEDI

AI/ML Developer
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LeetCode | Naukri Code360

PROFESSIONAL SUMMARY

B.Tech CSE student (2026) with hands-on experience designing multi-agent AI systems using LangGraph and LangChain, including a StateGraph-based routing pipeline that cut LLM API costs by 30%, alongside a Retrieval-Augmented Generation (RAG) pipeline on PostgreSQL (pgvector) for semantic document search and retrieval. Strong Python and backend engineering foundation across full-stack (MERN, Next.js) and systems-level projects, with a solid grasp of DSA, OOPs, DBMS, and REST API design. Self-driven and fast at picking up new tools and frameworks.

SKILLS

- **AI / LLM Engineering:** LangGraph, LangChain, StateGraph, RAG, PostgreSQL/pgvector, Semantic Search, LLM Query Routing, Agentic Workflows
- **Programming Languages:** Python, C, C++, Java, JavaScript, TypeScript
- **CS Fundamentals:** Data Structures & Algorithms, OOPs, DBMS, Operating Systems, Computer Networks, Multithreading & Concurrency
- **Tools & Frameworks:** Node.js, Express.js, MongoDB, PostgreSQL, SQLite, React, Next.js, Firebase, GCP, Git/GitHub, Postman, REST APIs, FFmpeg, PySide6 (Qt6), uv
- **Coding Profiles:** 100+ DSA problems solved across LeetCode and Naukri Code360

WORK EXPERIENCE

Healthyonegram — IT Intern *Jan 2026 – 29 Jun 2026*
B1GCRM (Self-Driven Enterprise Architecture Project) — React, Node.js, PostgreSQL, LangGraph, LangChain

- **Agentic Orchestration:** Architected a StateGraph-based multi-agent routing system with LangGraph and LangChain, cutting synthesis prompt token size by 45% and LLM API costs by 30%.
- **Query Triage:** Built a Triage Router directing client queries to specialist agents (Support RAG, Sales, SQL Order Tracker) with 98% routing accuracy, eliminating order-status hallucinations via direct PostgreSQL retrieval.
- **Resiliency:** Engineered a self-healing multi-agent architecture with try/catch fallback boundaries and a 5ms fallback bridge, ensuring 100% message delivery reliability.
- **Context Optimization:** Tuned conversation memory with a sliding 5-message window and 800-character truncation guard, preventing token overflow while preserving order metadata.
- Developed a RAG knowledge base using PostgreSQL pgvector for secure, AI-powered semantic document search and retrieval.
- Architected a multi-tenant, omni-channel CRM platform (React, Node.js, PostgreSQL, Socket.io) with real-time WhatsApp/Instagram integrations, contributing 208K+ lines of code across the full stack.
- Built a visual automation flow builder (React Flow, Zustand); optimized backend with raw SQL and indexing, cutting campaign execution time by 90%; implemented RBAC and shipped 30+ automated integration tests.

BogEcom — MERN, Next.js, Firebase, GCP

- Developed and deployed a production e-commerce platform with 5,000+ MAU; integrated 2 payment gateways with JWT auth and RBAC, improving API/DB performance by 40%.
- Automated CI/CD with GitHub Actions, cutting deployment time by 80% and resolving 100+ production/testing issues.

PROJECTS

MediaForge AI — Automated Media Transcoding Daemon — Python, FFmpeg, SQLite, PySide6 (Qt6), Unix Domain Sockets

- Independently designed a thread-isolated background daemon with a filesystem watchdog, SQLite-backed FIFO job queue, Unix Domain Socket IPC, and a worker-thread pool for concurrent, non-blocking processing.
- Built wrapper classes with automated GPU detection/CPU fallback and SHA256-based duplicate detection (O(1) lookups); stress-tested against 100 concurrent files, peak RAM under 73MB.

Data Structure Visualizer — VS Code Extension — TypeScript, Tree-sitter, Babel, Mermaid.js, D3.js

- Built a full-featured extension for real-time visualization of arrays, trees, and linked lists with step-by-step algorithm animation, multi-language AST-based parsing, variable tracking, and breakpoints; achieved 92.8% test success across 69 automated tests.

Two-Way Sync: Airtable ↔ Trello — Python, REST APIs

- Built an automated synchronization engine eliminating 80–90% of manual updates by auto-creating and updating Trello tasks from Airtable leads via REST APIs.
- Designed idempotent logic ensuring 0 duplicate cards, with auto-healing flows to recreate deleted Trello cards, maintaining 100% data consistency and cutting sync errors by 70%+.

Summitly — Full-Stack Event Management Platform — MERN, React + Vite

- Built a full-stack event platform supporting 20+ events with secure JWT login across 10+ dynamic pages and a ticketing system managing 100+ ticket records, backed by scalable REST APIs across 3 core data models.

EDUCATION

JECRC University, Jaipur — B.Tech, Computer Science & Engineering — CGPA: 7.3 *2022 – 2026*
Central Public Sr. Sec. School, Udaipur — Class XII (Science) — 82% *2022*
Central Public Sr. Sec. School, Udaipur — Class X — 87% *2020*

CERTIFICATES

- Leadership and Team Effectiveness — NPTEL, Jan–Apr 2025 (Certificate ID: NPTEL25MG38S114740245)
- IBM Python 101 for Data Science — Jul 2025 (Certificate ID: f789e8af3738466393e53ab329fd8373)